

Agent360: A Concealment-First Bilateral Negotiator for ANL 2026

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Abstract

We present **Agent360**, a bilateral negotiator submitted to the Automated Negotiation League (ANL) 2026. The competition scores each agent on **Advantage** (deal quality relative to a reserved value) and **Concealing** (how poorly the opponent learns our true preferences from our bid stream). Our agent design follows a *concealment-first* principle: early offers deliberately misrepresent issue priorities so that frequency-based opponent learners infer an incorrect utility function, while later phases and acceptance rules extract profitable agreements. The agent uses no oracle access to the opponent’s class or true utility—only observed offers, negotiation seat, and the relative deadline time. We relate the bidding persona to theories of signaling, strategic misrepresentation, and the limits of honesty in non-cooperative negotiation.

1 Introduction

In bilateral alternating-offer negotiation, each proposal is both an economic act and a *signal* about private preferences. The ANL 2026 league makes this dual role explicit: an agent is rewarded not only for closing good deals, but also for preventing the opponent from building an accurate model of its true utility function. Formally, the league score is

$$\text{Score} = \text{Advantage} + \text{Concealing}, \tag{1}$$

where concealment is derived from the rank correlation between the agent’s true utility and the utility function the opponent inferred from the negotiation transcript.

Most student agents in the league learn from offers using Smith-style frequency models or similar adaptive estimators. Against such opponents, bidding one’s sincere top outcomes early tends to maximize short-term deal quality but minimizes concealing, because the opponent’s learner quickly converges on the correct issue weights. Agent360, therefore, treats preference *concealment* as a first-class design goal and tunes deal extraction only in ways that do not collapse the decoy persona.

This report describes the agent, its psychological motivation, its three-layer architecture, and the evaluation process that produced the final parameterization.

2 Background: Deception, Signaling, and Learning

2.1 Offers as preference signals

In multi-issue negotiation, an offer is a vector of issue values. Rational opponents treat repeated offers as evidence about the values the sender prefers. Smith-style frequency models count how often each value appears in the opponent’s bids, implementing a simple form of *preference elicitation from behavior*. If our early bids consistently favor values that are actually low-priority for us, the opponent’s estimated utility function shifts away from the truth, which directly increases concealment under the league metric.

This is distinct from verbal deception: ANL agents cannot negotiate in natural language. All misdirection must be encoded in the *bid stream*, analogous to non-verbal signaling in human bargaining (consistent choices that suggest false priorities).

2.2 Truth mechanisms versus competitive misrepresentation

Mechanism-design results (e.g. strategy-proof auction formats) show that honesty can be a dominant strategy when the *rules* reward truthful reporting. The ANL scoring rule has the opposite incentive on the concealment axis: if the opponent models the agent correctly, our concealment score falls. We therefore operate in a non-cooperative setting closer to competitive signaling than to incentive-compatible mechanism design.

We initially considered a *reverse-psychology* persona: bid sincere high-utility outcomes early (as if “playing truth”), then shift toward misdirection, exploiting opponents that systematically discount our offers. Local experiments showed that this variant improved advantage against some time-based negotiators but reduced concealment against BOA, MAP, and MiCRO-style learners—the dominant opponent family in both our benchmarks and the live tournament. We rejected reverse-truth as the submission strategy and retained a gradual decoy-to-truth persona instead.

2.3 Opponent modeling without labels

Human negotiators form *theories of mind* about whether the partner is conceding, posturing, or mirroring prior bids. Agent360 approximates this with a lightweight behavioral classifier (**mirror**, **learner**, **deceptive**, **conceding**, **unknown**) derived from bid concentration, trajectory shape, issue-flip rate, and echo detection—not from the opponent’s declared agent class. This avoids brittle, hard-coded routing while still allowing mode-dependent aspiration slopes, closing weights and bait guards.

3 Agent Architecture

Agent360 extends the NegMAS SAOCallNegotiator protocol with three cooperating layers.

1. **Bidding persona** — what we offer: decoy pool construction, phased transition, and closing bids.
2. **Opponent model** — what we infer: blended Smith estimators, trajectory features, and mode classification.
3. **Deal extraction** — when we accept: time and mode-dependent aspiration, bait detection, and deadline safeguards.

On each step with a partner offer, the agent updates all estimators, evaluates acceptance, and otherwise samples the next bid from the current phase. The first negotiator (seat 0) follows a stricter concealment profile because opening agents expose more of their bid stream to the opponent’s learner.

4 Concealing Bidding Strategy

Relative time $t \in [0, 1]$ divides the negotiation into three phases.

Decoy phase. We bid from a **maximal-mismatch decoy pool**: rational outcomes that disagree with our inferred true top-issue values on at least half of the issues, subject to a utility floor. When we open, transition is blocked until the opponent has made at least four offers reducing early leakage to curve-fitting learners. Decoy selection avoids recent repeats, breaks mirror tit-for-tat by perturbing echoed bids, and prefers offers with large utility jumps relative to our last bid. The first seat extends the decoy window to $t < 0.40$ (versus 0.35 for the second seat).

Transition phase ($t < 0.75$). We mix decoy outcomes with a widening band of true high-utility outcomes. The first negotiator keeps decoy mixing for most of the transition window (`FIRST_TRANSITION_DECOY_MIX_UNT`: 0.85) and reveals true preferences more slowly (`FIRST_TRANSITION_PROGRESS_SCALE` = 0.72). If the opponent is classified as conceding and the trajectory shows honest concession, we allow an *early exit* from decoy even before the usual time gate—trading a small amount of concealing for advantage when the partner is unlikely to exploit the signal.

Closing phase ($t \geq 0.75$). We sample rational outcomes and score each candidate by a blend of normalized self-utility and estimated opponent utility. The opponent weight increases through closing but is capped by mode (e.g. higher caps against learners and conceders, lower against deceptive or mirroring opponents). Against deceptive late bait-switching, inflated Smith scores are discounted when selecting closing bids.

Together, these mechanisms implement a *gradual preference revelation* strategy: the opponent’s learner is fed a sustained false early signal, then gradually corrected only when deal extraction dominates further concealment.

5 Opponent Modeling

All opponent inference uses only the observed bid stream, relative time, and negotiation seat. We never read the opponent’s agent class, true utility function, or internal state. The modeling stack has three roles: estimate what the opponent values, classify how they are negotiating, and publish a blended utility function that downstream bidding and acceptance logic can query.

5.1 Smith-style frequency estimation

The base estimator is a **Smith frequency model**: for each issue, we count how often the opponent chose each value in past offers. Values repeated often are treated as more valuable to them; per-issue scores are normalized and averaged into a scalar utility for any outcome. This matches the learning bias of many league agents (BOA, MAP, MiCRO) and provides a stable prior over the full negotiation.

We maintain three refinements of this idea:

- **Recency-blended Smith** (second seat): when at least two recent offers exist in a window of five bids, we blend full-history Smith with recent-only Smith. Weight on the recent estimate grows with sample count (up to ≈ 0.68), so we track learners that adapt quickly after our persona shifts.
- **Time-weighted Smith**: bids after $t \geq 0.40$ are counted three times when building effective frequencies. Late offers often reflect genuine priorities after decoy posturing ends; up-weighting them reduces contamination from early misleading bids—both theirs and ours.

- **Issue-weighted Smith:** issues where the opponent repeats the same value (low spread) receive lower weight; issues with diverse values receive higher weight. This filters decoy noise on “frozen” issues while emphasizing dimensions where real trade-offs appear.

5.2 Trajectory and concession dynamics

The **offer trajectory model** records pairs $(t, \hat{u}_{\text{Smith}})$ at each opponent bid and treats the stream as a time series of revealed preference strength. From it we compute:

- **Concession slope** — linear trend of Smith utility over time; slopes below ≈ -0.04 indicate honest monotonic concession.
- **Non-monotonicity** — two or more significant utility *increases* suggest strategic feints or bait-and-switch rather than steady concession.
- **Predicted utility at t** — extrapolation used to test whether a late offer is inconsistent with the opponent’s prior trajectory (a bluff signal).

Trajectory features feed both mode classification and bait rejection at acceptance time.

5.3 Behavioral mode classification

Rather than routing by opponent *type* (which is unknown in competition), we classify *behavior* into five modes: **mirror**, **learner**, **deceptive**, **conceding**, and **unknown**. The decision tree uses mirror echo detection first (at least three matches to our recent bids in a window of four opponent offers), then trajectory sample count, Smith concentration, concession slope, and concealment-tactic heuristics.

Deceptive signals. We label an opponent deceptive when their bid stream shows concealment tactics, for example: identical early offers, high early issue-flip rate (≥ 0.25), wide early Smith spread (≥ 0.12), non-monotone utility paths, or late bait-switching (preferred values flip on many issues after $t \geq 0.40$). Plain BOA/MAP-style learners usually fall into **learner**, not **deceptive**, which limits false bait triggers against honest frequency learners.

Table 1 summarizes how each classified mode retunes closing, acceptance, Smith blending, and persona behavior. Classification is descriptive of *behavior*, not opponent class name.

Table 1: Classified opponent mode and agent response (V4.2).

Mode	Cap	Slope	Smith	Bait	Notes
Conceding	0.52	0.42	Full blend	No	Early decoy exit
Learner	0.48	0.52	Full blend	No	Self-utility boost late
Deceptive	0.32	0.58	Discounted	Yes	Bait reject active
Mirror	0.38	0.55	Plain Smith	No	Anti-mirror bids
Unknown	0.40	0.55	Full blend	No	Conservative de- fault

Mode-dependent control. The closing cap limits how much opponent utility weights closing bids; the aspiration slope controls how quickly we lower our acceptance threshold over time. Against deceptive opponents, inflated Smith estimates are discounted before bid scoring and late offers may be rejected as bait when they diverge from the trajectory prediction. Conceding opponents may trigger early decoy exit when concession slope is honestly negative; mirror opponents skip adaptive Smith blends so tit-for-tat echoes do not corrupt the published model.

5.4 Published opponent utility blend

The league measures Concealing from how well the *opponent* models *us*; symmetrically, we publish `opponent_ufun` in private information so the opponent can score Advantage. The published estimate is built in stages:

1. Start from full-history Smith.
2. If enough late bids exist ($t \geq 0.40$), blend toward a late-phase Smith estimate; blend weight increases with late sample count (cap ≈ 0.55 – 0.68 , higher when we open).
3. If at least three late offers exist, further blend toward issue-weighted late Smith.
4. On the second seat, blend toward recency Smith when recent data exist; for learner/conceding/unknown modes, add a stable-issue match term when recency is strong.
5. If mode is `mirror`, skip adaptive blends and return plain Smith.
6. Clamp all values to $[0, 1]$.

Closing bid selection and bait detection query this pipeline; against deceptive opponents, selected Smith utilities may be discounted before scoring candidate agreements.

5.5 Anti-mirror response

When the opponent repeats our last bid, mirror detection fires and we break synchrony by perturbing the next offer from an anti-mirror pool (small random change on one issue). Tit-for-tat mirroring otherwise freezes preference learning on both sides and depresses Concealing without improving Advantage.

6 Acceptance Strategy

The agent accepts a partner’s offer when any of the following holds:

- **Catastrophe floor:** $t \geq 0.95$ and utility $\geq$ reserved value.
- **Mode aspiration:** utility $\geq u_{\max}(1 - \text{slope} \cdot t)$ and above reserved value; the aspiration slope depends on the classified opponent mode.
- **AC-next:** offer utility is at least that of our next concealing bid.
- **Deadline safe:** $t > 0.90$ and utility above reserved value.

Before the deadline-safe window ($t \leq 0.90$), offers that pass aspiration may still be rejected if they appear to be **bait**: the opponent is classified as deceptive, shows concealment tactics, and the Smith-estimated utility exceeds the trajectory prediction by more than a threshold (`ACCEPT_BAIT_THRESHOLD` = 0.14).

During development, we tested an *escape tier* that accepted above reserved value from $t \approx 0.88$ onward, intended to reduce timeout penalties. Although this improved some local stress tests, tournament and proxy runs showed that it accepted too many mediocre or deceptive agreements against the real opponent mix. Currently, the agent deliberately disables mid-game escape acceptance and relies on catastrophe accept only near timeout.

7 Evaluation and Development Process

We evaluated each candidate agent on a repeatable local panel before submission:

- **Learners:** BOA, MAP, MiCRO across competition scenarios (both seats).
- **Stress:** time-based NegMAS negotiators (Boulware, Conceder, Tough, etc.).
- **Sparring:** in-house deceptive opponents (mirror, bait-switch, strong learner variants).

Each matchup cell is repeated with fixed step counts; we report mean Advantage, Concealing, and their sum.

Table 2 summarizes a representative development snapshot on the combined panel. Official tournament runs use a different opponent mix and panel size; we therefore treated leaderboard feedback as the final gate for acceptance and closing changes, not proxy means alone.

Table 2: Local proxy panel (mean Score per matchup cell; development snapshot).

Opponent group	Advantage	Concealing	Score
Learners (BOA/MAP/MiCRO)	0.56	0.71	1.27
Stress (time-based)	0.53	0.90	1.43
Deceptive sparring	0.61	0.58	1.19
Panel mean	0.55	0.79	1.34

Iterative refinement. The path from an early submission (strong concealing but weak advantage against learners) to the current Agent360 added: conceding-triggered early decoy exit, relaxed bait-discount thresholds, mode-dependent closing caps, first-seat decoy rotation, anti-mirror bid pools, and catastrophe acceptance near timeout. Changes that improved isolated local cells but hurt the broader distribution—stricter first-seat gates, shorter second-seat decoy windows, mid-game escape acceptance, and stall-based accept rules—were removed after broader evaluation showed regressions.

8 Discussion and Conclusions

Agent360 is a concealment-first bilateral negotiator grounded in signaling theory and empirical tests of alternative personas. Its core idea is psychologically motivated but implemented entirely through bid streams: early maximal-mismatch decoys mislead frequency learners, gradual transition limits preference leakage, and mode-aware closing and acceptance extract Advantage without surrendering Concealing to deceptive or mirroring opponents.

Three lessons emerged from development. First, **concealing and advantage require separate controls**; strengthening acceptance without preserving the decoy persona hurts the combined score against learner-heavy fields. Second, **evaluation must span multiple opponent families**; learner, stress, and deceptive sparring panels together catch failures that no single benchmark family reveals.

alone. Third, **seat asymmetry matters**; first and second negotiators see different information, and asymmetric decoy length and min-offer gates improved learner matchups without oracle routing.

The submitted agent operates without privileged information, adapts to inferred opponent behavior, and balances strategic misrepresentation of preferences with profitable agreement formation.